

# Phase 3:

# Problem Selection

## Overview

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In this phase, you will select a concrete medical image classification problem from a recent, high-quality research paper and conduct a focused investigation into its data, preprocessing choices, modeling assumptions, and experimental setup. You will now explore what makes a problem meaningful, challenging, and suitable for deep learning in real-world medical settings.

You will:

- Choose a peer-reviewed paper (published in 2021 or later) from a top-tier venue (Q1 journal or major conference like CVPR, MICCAI, NeurIPS, etc.) that includes public code and dataset access.
- Analyze the clinical motivation, dataset characteristics, class semantics, and design choices (especially preprocessing and augmentation).
- Justify why deep learning is appropriate for the task and how symmetry, normalization, and data constraints shape model behavior.

This phase emphasizes critical thinking, domain awareness, and reproducibility—key skills for responsible AI in healthcare.

## Step 1: Problem Selection & Justification (40%)

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1. Select a paper that satisfies as many of the following criteria as possible:
  - a. Published in 2021 or later.
  - b. Appears in a Q1 journal (e.g., IEEE TMI, Medical Image Analysis) or a top-tier conference (e.g., CVPR, ICCV, MICCAI, NeurIPS).
  - c. Provides publicly available code and clear instructions to load or download the dataset.
  - d. Includes fair comparisons with prior work.
  - e. Has at least one author with a demonstrated track record in medical imaging or vision (e.g., multiple relevant publications).
2. Justify your chosen problem by answering the following:
  - a. Why does this problem matter? (Clinical impact, prevalence, diagnostic challenges, etc.)

- b. Why is deep learning suitable here? (Limitations of traditional methods, need for representation learning, etc.)
- c. How was the data collected? (Source, labeling protocol, patient demographics, ethical considerations.)
- d. What do the classes represent? (e.g., disease subtypes, severity levels, anatomical regions). Clarify whether the task is multi-class, multi-label, or binary.

## Step 2: Preprocessing & Augmentation Analysis (60%)

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1. Describe and implement the preprocessing pipeline used in the paper (or a reasonable approximation if not fully specified). For each step, explain:
  - a. What it does (e.g., resize, normalization, filtering).
  - b. Why it matters (e.g., normalization stabilizes training; resizing enables compatibility with pretrained backbones).
2. Analyze the augmentations used (or propose medically plausible ones if none are specified). For each augmentation, explain:
  - What transformation is applied (e.g., random horizontal flip, rotation  $\pm 15^\circ$ ).
  - What symmetry or invariance it implicitly encodes (e.g., flip assumes left-right symmetry; rotation assumes orientation invariance).
  - Whether it is clinically valid (e.g., flipping a retinal scan may be invalid due to anatomical asymmetry).

 **Caution:** In medical imaging, not all standard augmentations are appropriate. Always justify choices based on domain knowledge.

3. Load the dataset and include visual examples (before/after) of key preprocessing and augmentation steps in your notebook.

## Submission Instruction

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The students should upload their report/notebook there according to this structure on their repository in <https://github.com/UT-AI-Lab>. The name of the submission file for this phase should be "*<Student Id>\_Phase3*".

```
|— student_name_1/
| |— "<Student Id>_Phase3".ipynb    # Main Jupyter notebook (answers + results)
| ...
```

 **Deadline for submission: 16 Aban.**

## Appendix: Key Concepts

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### Subject-Dependent vs. Subject-Independent Evaluation

- Subject-dependent: Train and test sets may contain images from the same patient. This often overestimates performance.
- Subject-independent: All images from a given patient appear only in train or test, never both. This better reflects real-world deployment and is strongly preferred in medical imaging.

### The Role of Symmetry in Augmentation

Augmentations are not just “data tricks”—they encode inductive biases. For example:

- Horizontal flip → assumes bilateral symmetry (valid for skin lesions, questionable for brain MRI).
- Rotation → assumes orientation invariance (valid for histopathology patches, invalid for chest X-rays where anatomy is fixed).

Choosing augmentations wisely ensures your model learns medically meaningful invariances, not artifacts.

### How to Verify if a Journal is Q1

To determine whether a journal belongs to Quartile 1 (Q1), you can use the SCImago Journal & Country Rank (SJR) portal: <https://www.scimagojr.com>.

Steps:

1. Go to <https://www.scimagojr.com>.
2. Search for the journal name (e.g., IEEE Transactions on Medical Imaging).
3. In the results, look at the “Quartile” column under the relevant subject category (e.g., Radiology, Nuclear Medicine & Medical Imaging or Computer Vision and Pattern Recognition).

If the journal is listed as Q1 in at least one relevant category, it satisfies the Q1 requirement for this phase. For conferences, quartile rankings do not apply. Instead, consider them acceptable if they are widely recognized as top-tier venues in computer vision, medical image analysis, or AI.